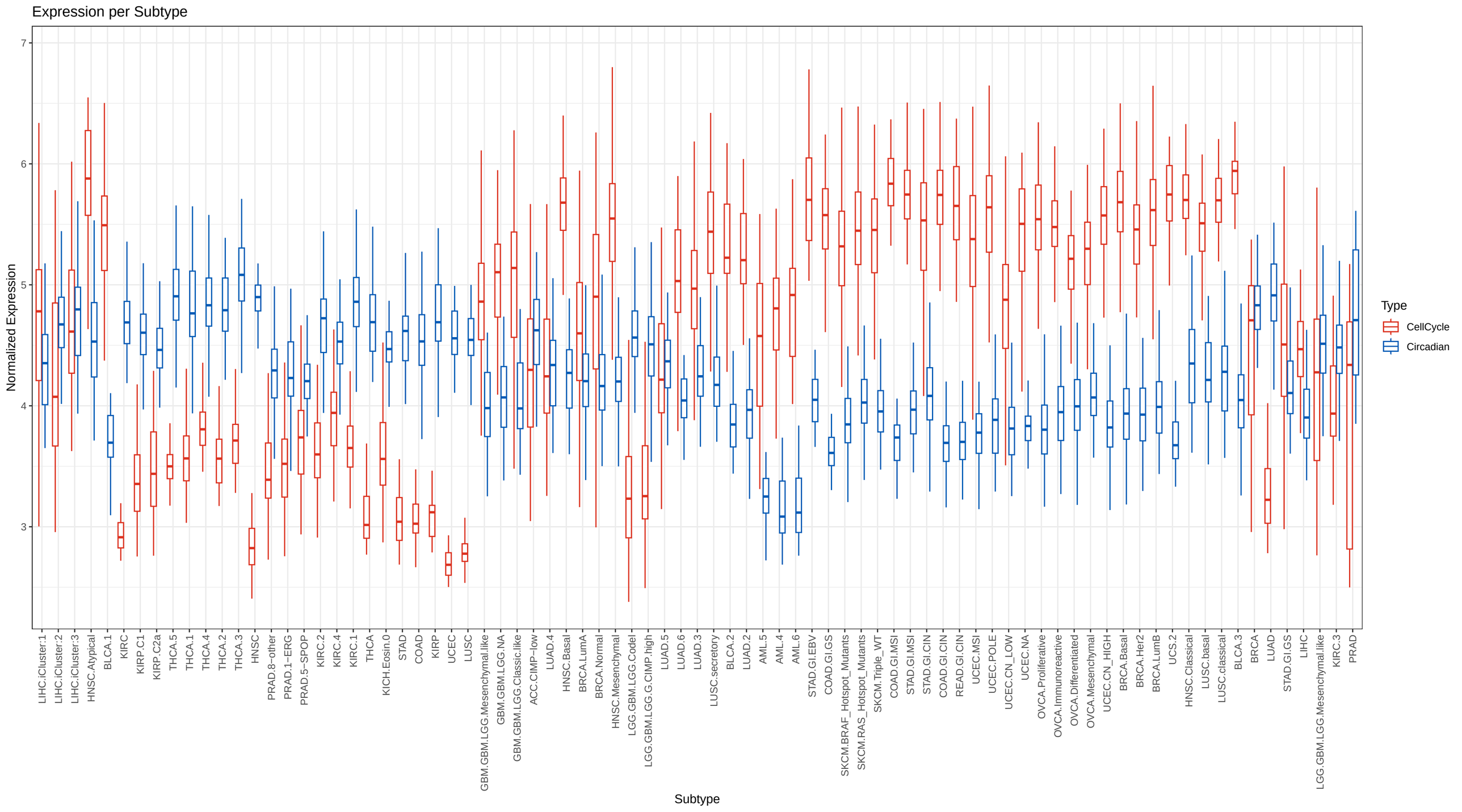
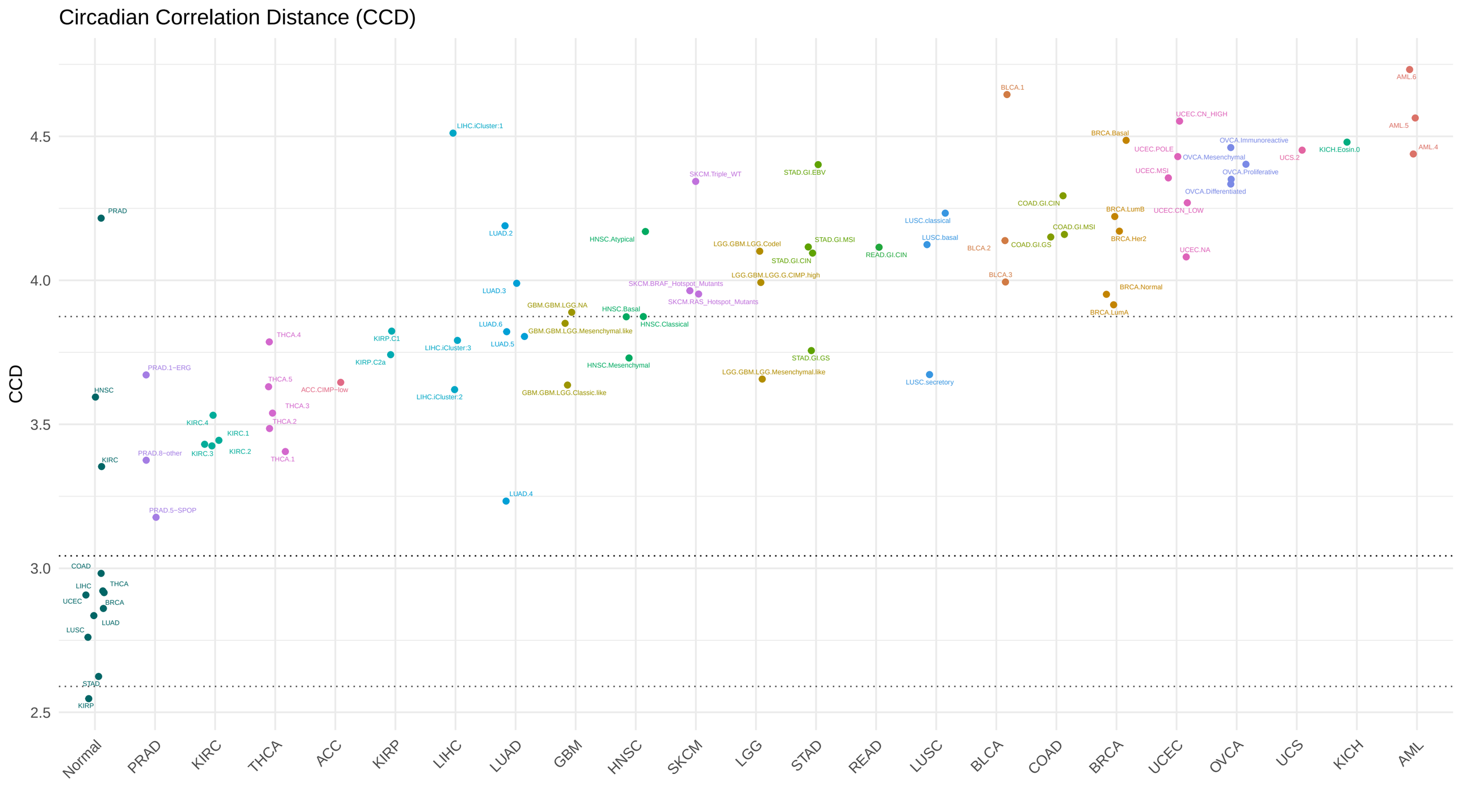
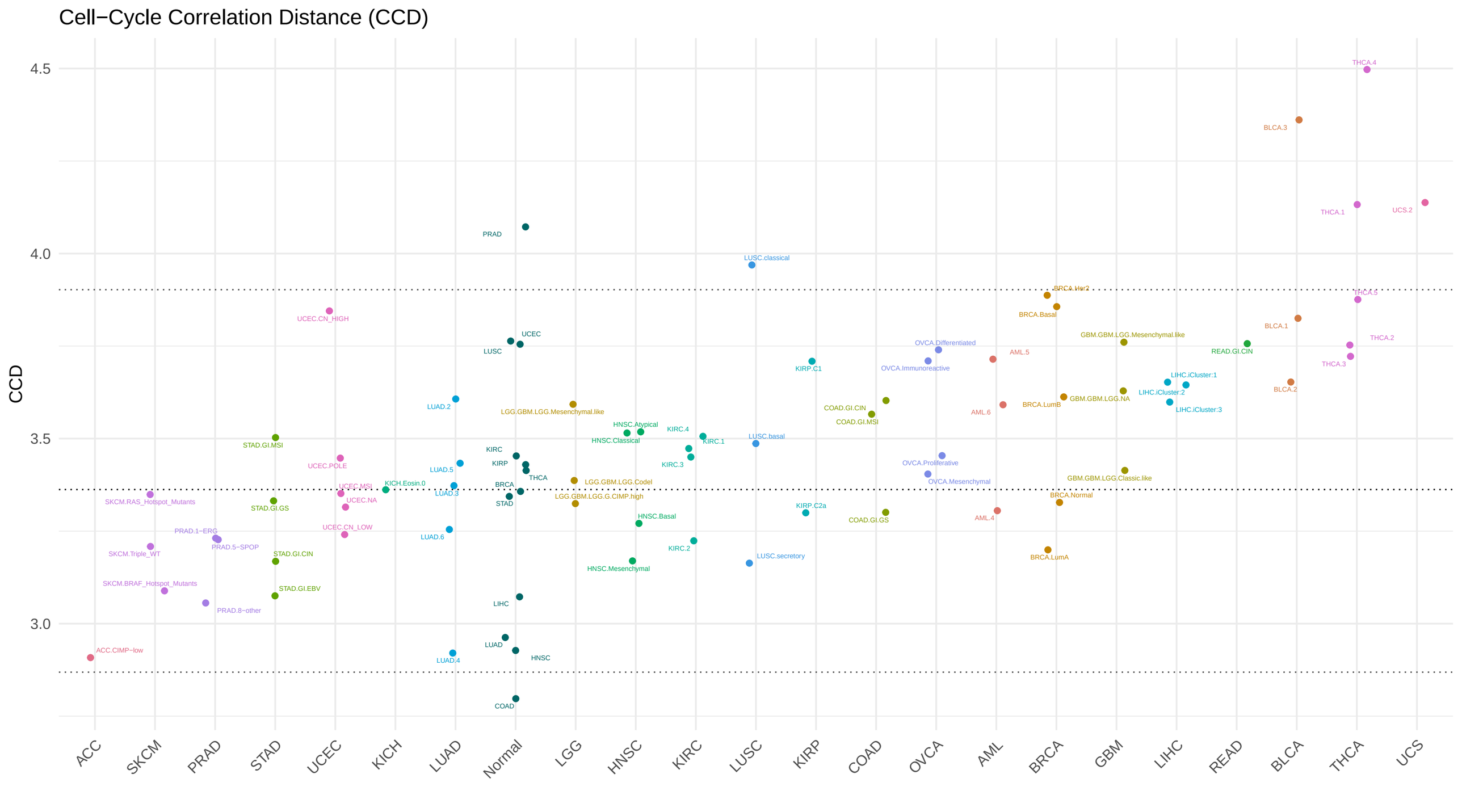
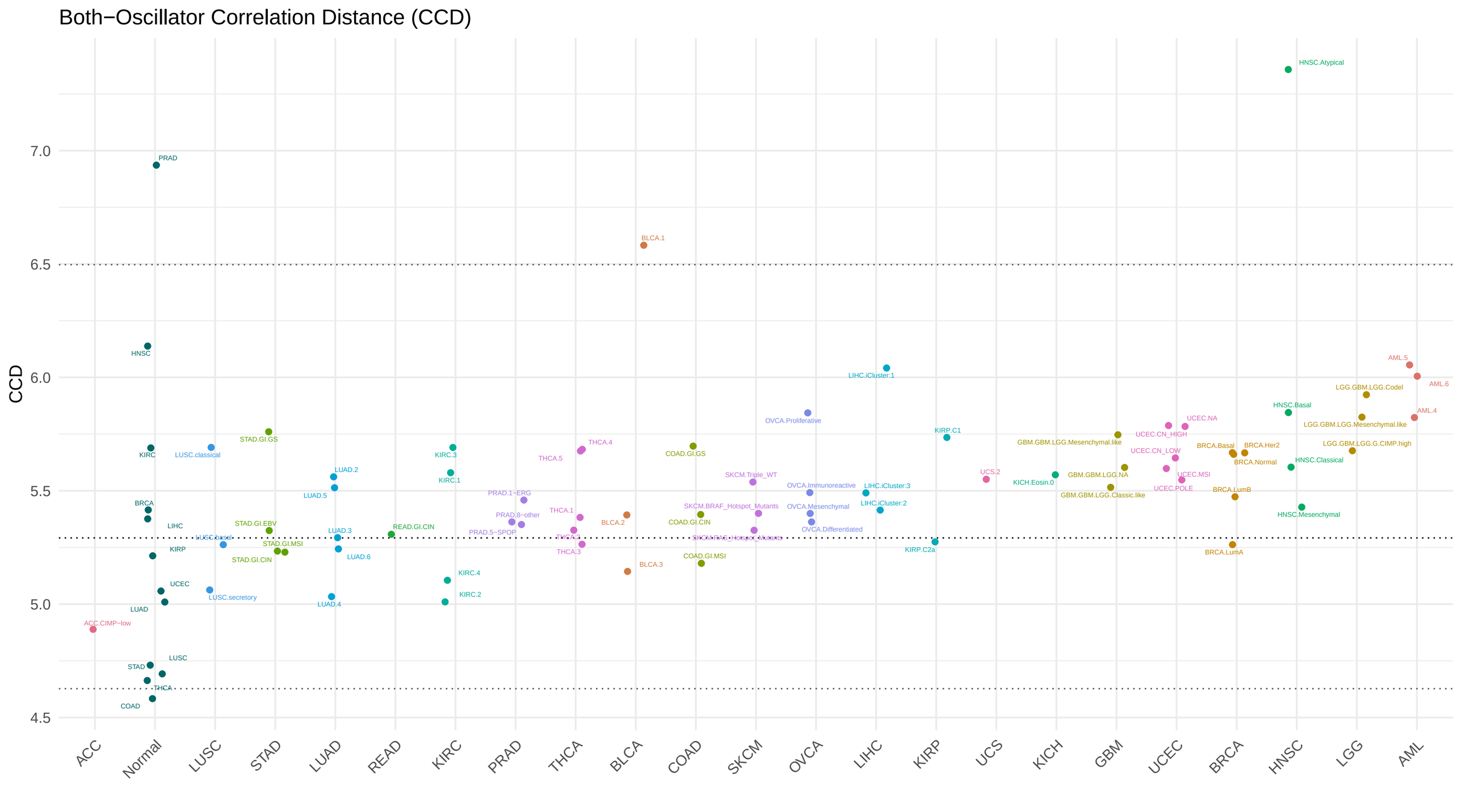
4
2
3
5
1
0.140
−0.572
0.081
0.485
−0.194
2.Circadian Clock
1.Both Oscillators
3.Cell Cycle
Spearman's rho
1
0.5
0
−0.5
−1
Sample Type
Cancer
Normal
log2(cell cycle / circadian)
Negative
Positive
SampleType
LIHC.iCluster:1
LIHC.iCluster:2
LIHC.iCluster:3
HNSC.Atypical
BLCA.1
KIRC
KIRP.C1
KIRP.C2a
THCA.5
THCA.1
THCA.4
THCA.2
THCA.3
HNSC
PRAD.8−other
PRAD.1−ERG
PRAD.5−SPOP
KIRC.2
KIRC.4
KIRC.1
THCA
KICH.Eosin.0
STAD
COAD
KIRP
UCEC
LUSC
GBM.GBM.LGG.Mesenchymal.like
GBM.GBM.LGG.NA
GBM.GBM.LGG.Classic.like
ACC.CIMP−low
LUAD.4
HNSC.Basal
BRCA.LumA
BRCA.Normal
HNSC.Mesenchymal
LGG.GBM.LGG.Codel
LGG.GBM.LGG.G.CIMP.high
LUAD.5
LUAD.6
LUAD.3
LUSC.secretory
BLCA.2
LUAD.2
AML.5
AML.4
AML.6
STAD.GI.EBV
COAD.GI.GS
SKCM.BRAF_Hotspot_Mutants
SKCM.RAS_Hotspot_Mutants
SKCM.Triple_WT
COAD.GI.MSI
STAD.GI.MSI
STAD.GI.CIN
COAD.GI.CIN
READ.GI.CIN
UCEC.MSI
UCEC.POLE
UCEC.CN_LOW
UCEC.NA
OVCA.Proliferative
OVCA.Immunoreactive
OVCA.Differentiated
OVCA.Mesenchymal
UCEC.CN_HIGH
BRCA.Basal
BRCA.Her2
BRCA.LumB
UCS.2
HNSC.Classical
LUSC.basal
LUSC.classical
BLCA.3
BRCA
LUAD
STAD.GI.GS
LIHC
GBM.GBM.LGG.Mesenchymal.like
KIRC.3
PRAD


Expression per Subtype
Normalized Expression
7
6
5
4
3
LIHC.iCluster:1
LIHC.iCluster:2
LIHC.iCluster:3
HNSC.Atypical
BLCA.1
KIRC
KIRP.C1
KIRP.C2a
THCA.5
THCA.1
THCA.4
THCA.2
THCA.3
HNSC
PRAD.8−other
PRAD.1−ERG
PRAD.5−SPOP
KIRC.2
KIRC.4
KIRC.1
THCA
KICH.Eosin.0
STAD
COAD
KIRP
UCEC
LUSC
GBM.GBM.LGG.Mesenchymal.like
GBM.GBM.LGG.NA
GBM.GBM.LGG.Classic.like
ACC.CIMP−low
LUAD.4
HNSC.Basal
BRCA.LumA
BRCA.Normal
HNSC.Mesenchymal
LGG.GBM.LGG.Codel
LGG.GBM.LGG.G.CIMP.high
LUAD.5
LUAD.6
LUAD.3
LUSC.secretory
BLCA.2
LUAD.2
AML.5
AML.4
AML.6
STAD.GI.EBV
COAD.GI.GS
SKCM.BRAF_Hotspot_Mutants
SKCM.RAS_Hotspot_Mutants
SKCM.Triple_WT
COAD.GI.MSI
STAD.GI.MSI
STAD.GI.CIN
COAD.GI.CIN
READ.GI.CIN
UCEC.MSI
UCEC.POLE
UCEC.CN_LOW
UCEC.NA
OVCA.Proliferative
OVCA.Immunoreactive
OVCA.Differentiated
OVCA.Mesenchymal
UCEC.CN_HIGH
BRCA.Basal
BRCA.Her2
BRCA.LumB
UCS.2
HNSC.Classical
LUSC.basal
LUSC.classical
BLCA.3
BRCA
LUAD
STAD.GI.GS
LIHC
LGG.GBM.LGG.Mesenchymal.like
KIRC.3
PRAD
Subtype
Type
CellCycle
Circadian

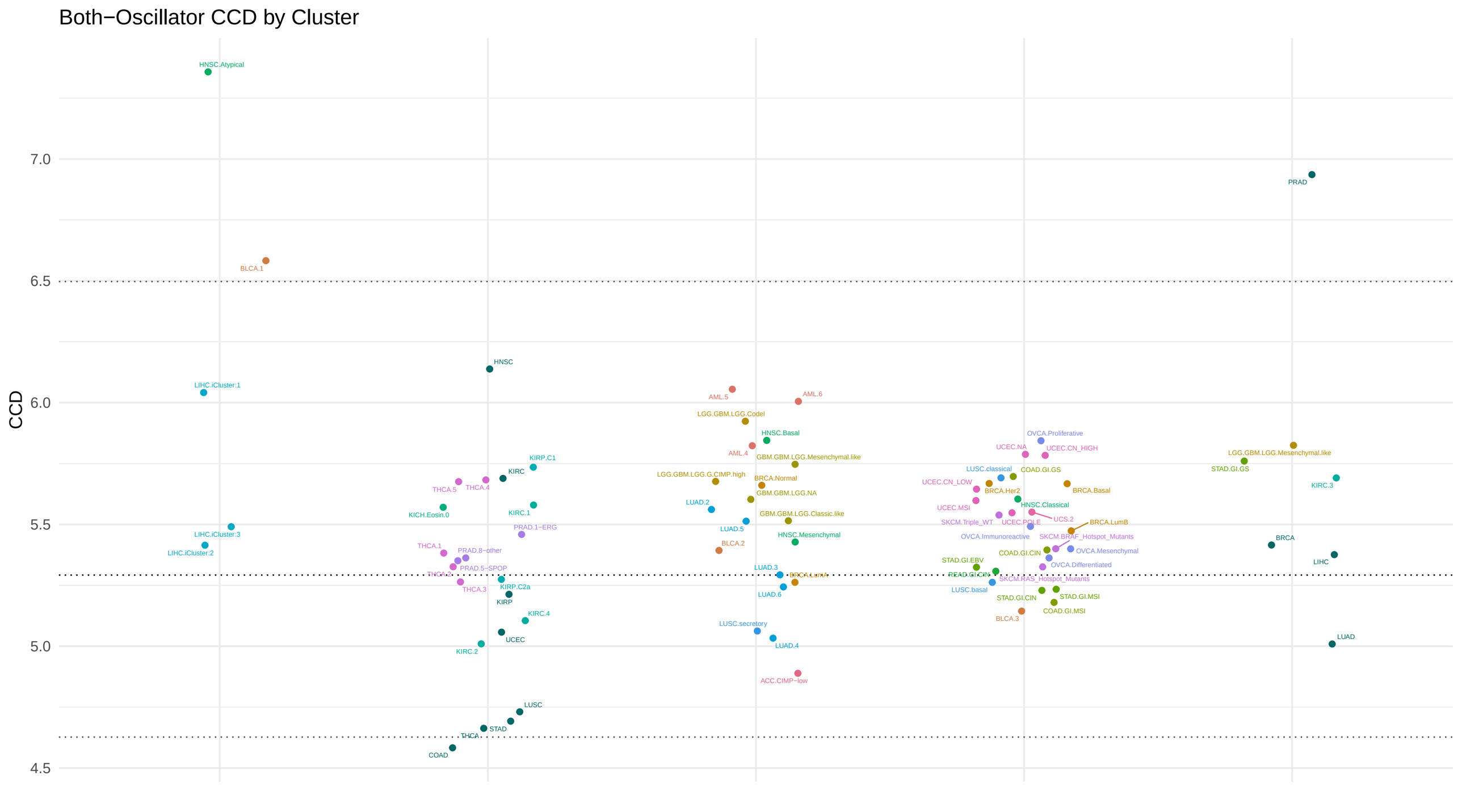

Circadian Correlation Distance (CCD)
CCD
4.5
4.0
3.5
3.0
2.5
Normal
PRAD
KIRC
THCA
ACC
KIRP
LIHC
LUAD
GBM
HNSC
SKCM
LGG
STAD
READ
LUSC
BLCA
COAD
BRCA
UCEC
OVCA
UCS
KICH
AML
LIHC.iCluster:1
PRAD
LUAD.2
HNSC.Atypical
SKCM.Triple_WT
STAD.GI.EBV
LGG.GBM.LGG.Codel
STAD.GI.MSI
STAD.GI.CIN
READ.GI.CIN
LUSC.classical
LUSC.basal
BLCA.1
BLCA.2
COAD.GI.CIN
COAD.GI.MSI
COAD.GI.GS
BRCA.Basal
BRCA.LumB
BRCA.Her2
UCEC.CN_HIGH
UCEC.POLE
UCEC.MSI
UCEC.CN_LOW
UCEC.NA
OVCA.Immunoreactive
OVCA.Mesenchymal
OVCA.Proliferative
OVCA.Differentiated
UCS.2
KICH.Eosin.0
AML.6
AML.5
AML.4
LUAD.3
LGG.GBM.LGG.G.CIMP.high
SKCM.BRAF_Hotspot_Mutants
SKCM.RAS_Hotspot_Mutants
BLCA.3
BRCA.Normal
BRCA.LumA
GBM.GBM.LGG.NA
HNSC.Basal
HNSC.Classical
LUAD.6
GBM.GBM.LGG.Mesenchymal.like
LUAD.5
THCA.4
KIRP.C1
LIHC.iCluster:3
KIRP.C2a
HNSC.Mesenchymal
STAD.GI.GS
LGG.GBM.LGG.Mesenchymal.like
LUSC.secretory
PRAD.1−ERG
THCA.5
ACC.CIMP−low
HNSC
THCA.3
KIRC.4
THCA.2
KIRC.1
KIRC.3
KIRC.2
PRAD.8−other
THCA.1
KIRC
LIHC.iCluster:2
GBM.GBM.LGG.Classic.like
LUAD.4
PRAD.5−SPOP
COAD
LIHC
THCA
UCEC
BRCA
LUAD
LUSC
STAD
KIRP

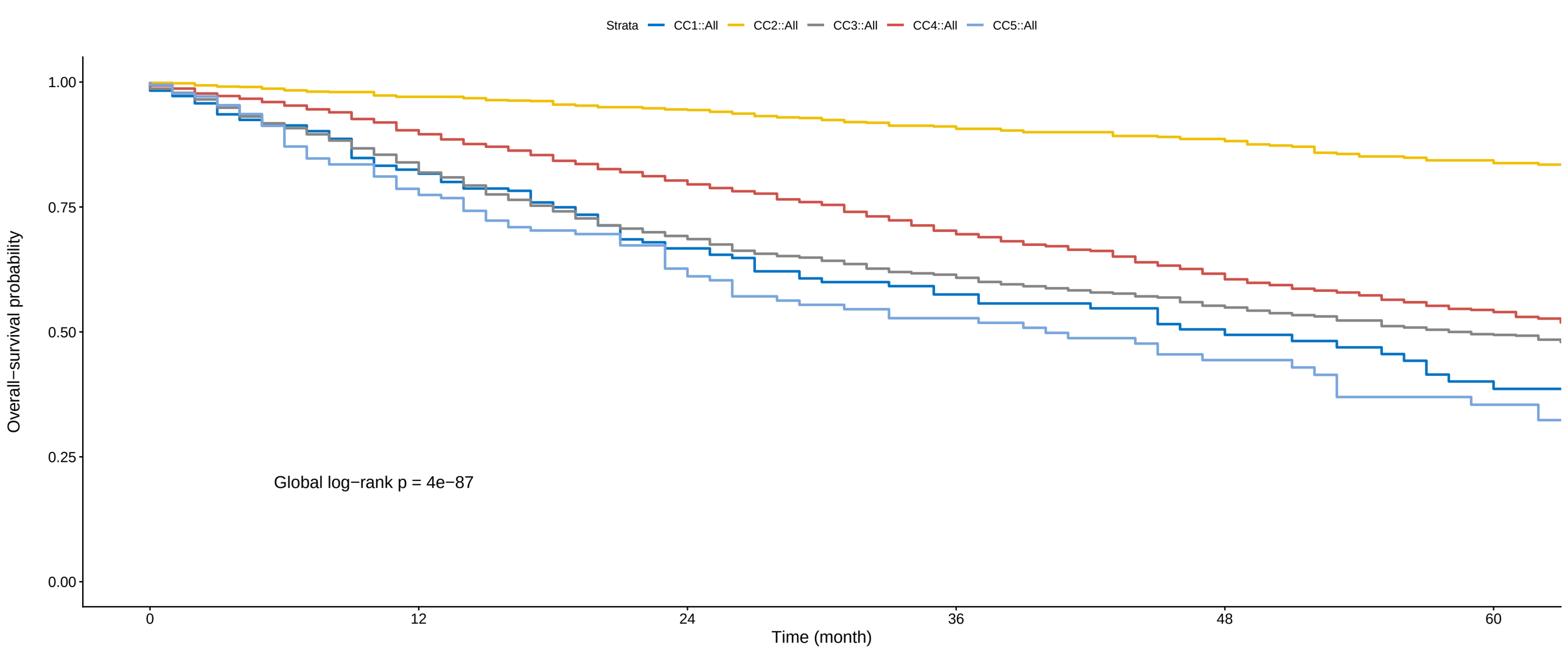

Cell−Cycle Correlation Distance (CCD)
CCD
4.5
4.0
3.5
3.0
ACC
SKCM
PRAD
STAD
UCEC
KICH
LUAD
Normal
LGG
HNSC
KIRC
LUSC
KIRP
COAD
OVCA
AML
BRCA
GBM
LIHC
READ
BLCA
THCA
UCS
ACC.CIMP−low
SKCM.BRAF_Hotspot_Mutants
SKCM.RAS_Hotspot_Mutants
SKCM.Triple_WT
PRAD.1−ERG
PRAD.5−SPOP
PRAD.8−other
PRAD
STAD.GI.CIN
STAD.GI.EBV
STAD.GI.GS
STAD.GI.MSI
UCEC.CN_HIGH
UCEC.CN_LOW
UCEC.MSI
UCEC.NA
UCEC.POLE
UCEC
KICH.Eosin.0
LUAD.2
LUAD.3
LUAD.4
LUAD.5
LUAD.6
LUAD
LUSC
LUSC.classical
LUSC.basal
LUSC.secretory
LIHC
HNSC
HNSC.Atypical
HNSC.Basal
HNSC.Classical
HNSC.Mesenchymal
LGG.GBM.LGG.Mesenchymal.like
LGG.GBM.LGG.Codel
LGG.GBM.LGG.G.CIMP.high
KIRC
KIRC.1
KIRC.2
KIRC.3
KIRC.4
KIRP
KIRP.C1
KIRP.C2a
BRCA
BRCA.Her2
BRCA.Basal
BRCA.LumB
BRCA.Normal
BRCA.LumA
STAD
THCA
COAD
COAD.GI.CIN
COAD.GI.MSI
COAD.GI.GS
OVCA.Differentiated
OVCA.Immunoreactive
OVCA.Proliferative
OVCA.Mesenchymal
AML
AML.5
AML.6
AML.4
GBM
GBM.GBM.LGG.Mesenchymal.like
GBM.GBM.LGG.NA
GBM.GBM.LGG.Classic.like
LIHC.iCluster:1
LIHC.iCluster:2
LIHC.iCluster:3
READ
READ.GI.CIN
BLCA
BLCA.3
BLCA.1
BLCA.2
THCA.4
THCA.5
THCA.1
THCA.2
THCA.3
UCS
UCS.2


Both−Oscillator Correlation Distance (CCD)
CCD
7.0
6.5
6.0
5.5
5.0
4.5
ACC
Normal
LUSC
STAD
LUAD
READ
KIRC
PRAD
THCA
BLCA
COAD
SKCM
OVCA
LIHC
KIRP
UCS
KICH
GBM
UCEC
BRCA
HNSC
LGG
AML
ACC.CIMP−low
BRCA
COAD
HNSC
KIRC
KIRP
LIHC
LUAD
LUSC
PRAD
STAD
THCA
UCEC
BLCA.1
BLCA.2
BLCA.3
COAD.GI.CIN
COAD.GI.GS
COAD.GI.MSI
GBM.GBM.LGG.Classic.like
GBM.GBM.LGG.Mesenchymal.like
GBM.GBM.LGG.NA
HNSC.Atypical
HNSC.Basal
HNSC.Classical
HNSC.Mesenchymal
KICH.Eosin.0
KIRC.1
KIRC.2
KIRC.3
KIRC.4
KIRP.C1
KIRP.C2a
LGG.GBM.LGG.Codel
LGG.GBM.LGG.G.CIMP.high
LGG.GBM.LGG.Mesenchymal.like
LIHC.iCluster:1
LIHC.iCluster:2
LIHC.iCluster:3
LUAD.2
LUAD.3
LUAD.4
LUAD.5
LUAD.6
LUSC.basal
LUSC.classical
LUSC.secretory
OVCA.Differentiated
OVCA.Immunoreactive
OVCA.Mesenchymal
OVCA.Proliferative
PRAD.1−ERG
PRAD.5−SPOP
PRAD.8−other
READ.GI.CIN
SKCM.BRAF_Hotspot_Mutants
SKCM.Triple_WT
STAD.GI.CIN
STAD.GI.EBV
STAD.GI.GS
STAD.GI.MSI
THCA.1
THCA.2
THCA.3
THCA.4
THCA.5
UCEC.CN_HIGH
UCEC.CN_LOW
UCEC.MSI
UCEC.NA
UCEC.POLE
UCS.2
BRCA.Basal
BRCA.Her2
BRCA.LumA
BRCA.LumB
BRCA.Normal
AML.4
AML.5
AML.6


Circadian CCD by Cluster
CCD
4.5
4.0
3.5
3.0
2.5
AML.6
BLCA.1
AML.5
UCEC.CN_HIGH
BRCA.Basal
LIHC.iCluster:1
KICH.Eosin.0
AML.4
OVCA.Immunoreactive
UCS.2
STAD.GI.EBV
UCEC.POLE
OVCA.Mesenchymal
UCEC.MSI
OVCA.Proliferative
SKCM.Triple_WT
OVCA.Differentiated
UCEC.CN_LOW
COAD.GI.CIN
BRCA.LumB
LUAD.2
LUSC.classical
PRAD
BRCA.Her2
COAD.GI.MSI
HNSC.Atypical
BLCA.2
LUSC.basal
STAD.GI.MSI
COAD.GI.GS
READ.GI.CIN
LGG.GBM.LGG.Codel
STAD.GI.CIN
UCEC.NA
LGG.GBM.LGG.G.CIMP.high
LUAD.3
SKCM.BRAF_Hotspot_Mutants
BRCA.Normal
BLCA.3
BRCA.LumA
SKCM.RAS_Hotspot_Mutants
GBM.GBM.LGG.NA
GBM.GBM.LGG.Mesenchymal.like
HNSC.Classical
KIRP.C1
HNSC.Basal
LIHC.iCluster:3
LUAD.6
LUAD.5
THCA.4
HNSC.Mesenchymal
STAD.GI.GS
KIRP.C2a
LUSC.secretory
LGG.GBM.LGG.Mesenchymal.like
PRAD.1−ERG
ACC.CIMP−low
THCA.5
GBM.GBM.LGG.Classic.like
LIHC.iCluster:2
HNSC
THCA.3
KIRC.4
KIRC.3
THCA.2
THCA.1
KIRC.2
KIRC.1
PRAD.8−other
KIRC
LUAD.4
PRAD.5−SPOP
COAD
LIHC
THCA
BRCA
UCEC
LUAD
LUSC
STAD
KIRP

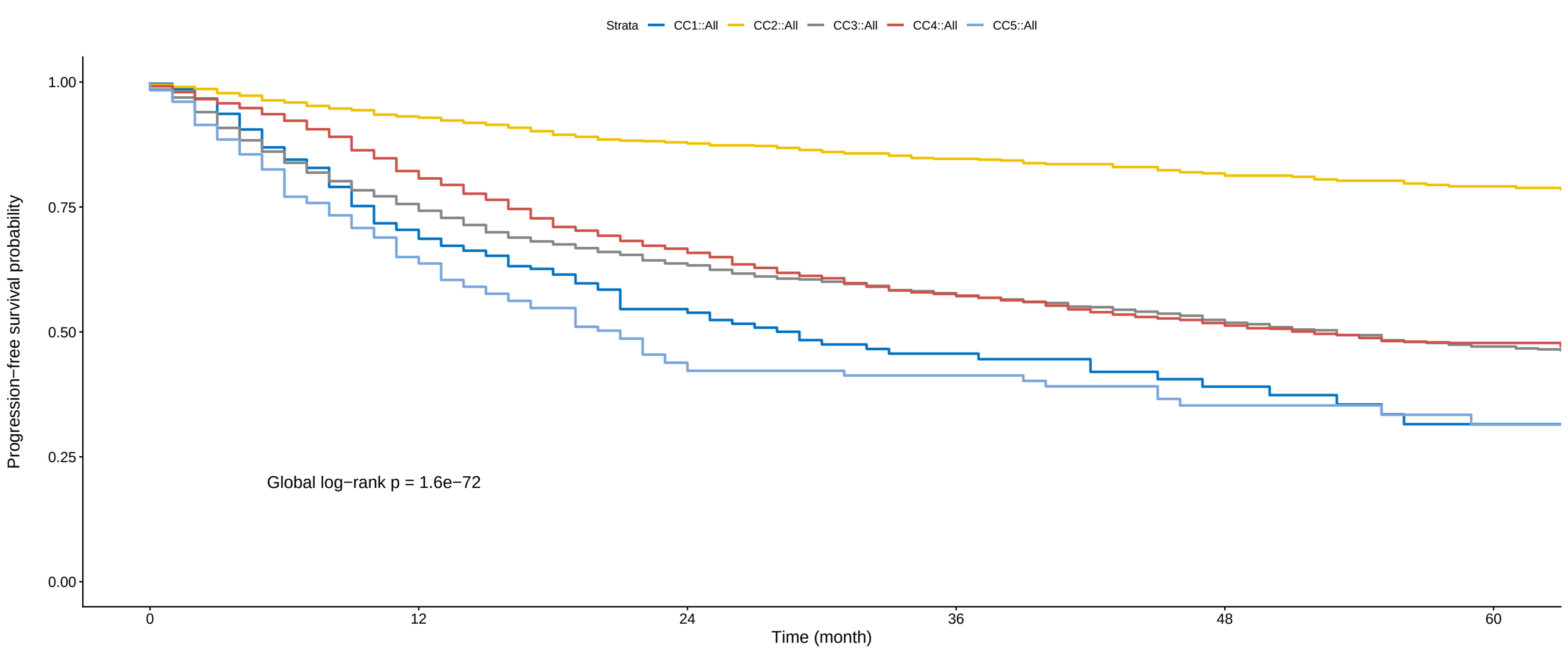

Cell−Cycle CCD by Cluster
CCD
4.5
4.0
3.5
3.0
THCA.4
BLCA.3
THCA.1
UCS.2
PRAD
LUSC.classical
THCA.5
BRCA.Her2
BRCA.Basal
BLCA.1
UCEC.CN_HIGH
THCA.2
UCEC
GBM.GBM.LGG.Mesenchymal.like
READ.GI.CIN
OVCA.Differentiated
THCA.3
LUSC
AML.5
OVCA.Immunoreactive
KIRP.C1
LIHC.iCluster:1
BLCA.2
LIHC.iCluster:2
GBM.GBM.LGG.NA
BRCA.LumB
LUAD.2
LGG.GBM.LGG.Mesenchymal.like
LIHC.iCluster:3
AML.6
COAD.GI.CIN
COAD.GI.MSI
HNSC.Classical
HNSC.Atypical
LUSC.basal
STAD.GI.MSI
KIRC.4
KIRC.3
KIRC.1
LUAD.5
KIRC
GBM.GBM.LGG.Classic.like
UCEC.POLE
OVCA.Proliferative
KIRP
LGG.GBM.LGG.Codel
OVCA.Mesenchymal
THCA
LUAD.3
BRCA
UCEC.MSI
SKCM.RAS_Hotspot_Mutants
STAD
KICH.Eosin.0
LGG.GBM.LGG.G.CIMP.high
BRCA.Normal
UCEC.NA
STAD.GI.GS
AML.4
KIRP.C2a
COAD.GI.GS
HNSC.Basal
UCEC.CN_LOW
PRAD.5−SPOP
PRAD.1−ERG
LUAD.6
SKCM.Triple_WT
KIRC.2
BRCA.LumA
HNSC.Mesenchymal
STAD.GI.CIN
LUSC.secretory
LIHC
STAD.GI.EBV
SKCM.BRAF_Hotspot_Mutants
PRAD.8−other
LUAD
HNSC
LUAD.4
ACC.CIMP−low
COAD


Both−Oscillator CCD by Cluster
CCD
7.0
6.5
6.0
5.5
5.0
4.5
HNSC.Atypical
BLCA.1
LIHC.iCluster:1
LIHC.iCluster:3
LIHC.iCluster:2
HNSC
KIRP.C1
KIRC
THCA.5
THCA.4
KICH.Eosin.0
KIRC.1
PRAD.1−ERG
THCA.1
PRAD.8−other
PRAD.5−SPOP
THCA.2
THCA.3
KIRP.C2a
KIRP
KIRC.4
UCEC
KIRC.2
LUSC
STAD
THCA
COAD
AML.5
AML.6
LGG.GBM.LGG.Codel
HNSC.Basal
AML.4
GBM.GBM.LGG.Mesenchymal.like
LGG.GBM.LGG.G.CIMP.high
BRCA.Normal
GBM.GBM.LGG.NA
LUAD.2
GBM.GBM.LGG.Classic.like
LUAD.5
HNSC.Mesenchymal
BLCA.2
LUAD.3
BRCA.LumA
LUAD.6
LUSC.secretory
LUAD.4
ACC.CIMP−low
OVCA.Proliferative
UCEC.NA
UCEC.CN_HIGH
LUSC.classical
COAD.GI.GS
UCEC.CN_LOW
BRCA.Her2
BRCA.Basal
UCEC.MSI
HNSC.Classical
SKCM.Triple_WT
UCEC.POLE
UCS.2
BRCA.LumB
OVCA.Immunoreactive
SKCM.BRAF_Hotspot_Mutants
COAD.GI.CIN
OVCA.Mesenchymal
STAD.GI.EBV
OVCA.Differentiated
READ.GI.CIN
SKCM.RAS_Hotspot_Mutants
LUSC.basal
STAD.GI.CIN
STAD.GI.MSI
COAD.GI.MSI
BLCA.3
PRAD
LGG.GBM.LGG.Mesenchymal.like
STAD.GI.GS
KIRC.3
BRCA
LIHC
LUAD



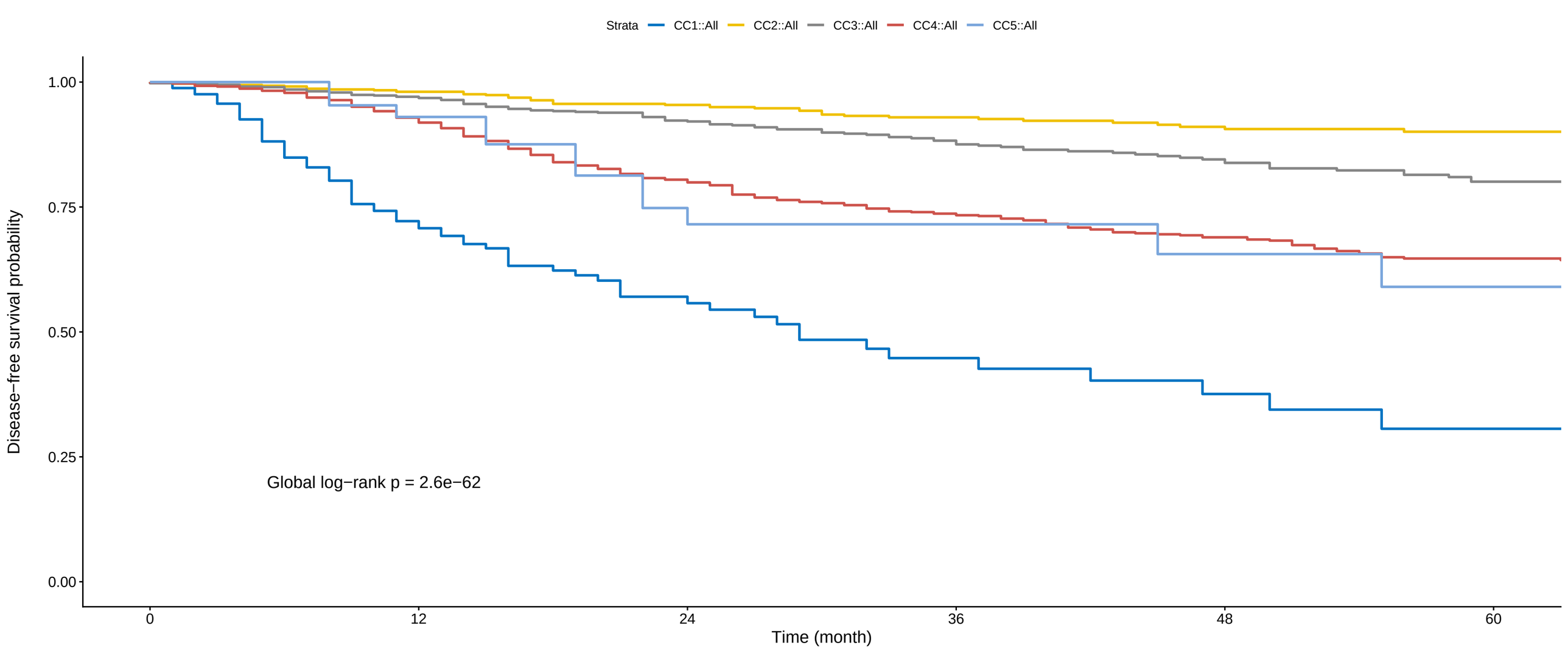
Number at risk

CC1::All	291	206	106	67	46	27
CC2::All	1243	1096	819	579	418	304
CC3::All	2201	1609	986	683	454	331
CC4::All	2746	2163	1448	984	711	513
CC5::All	182	128	81	57	36	23



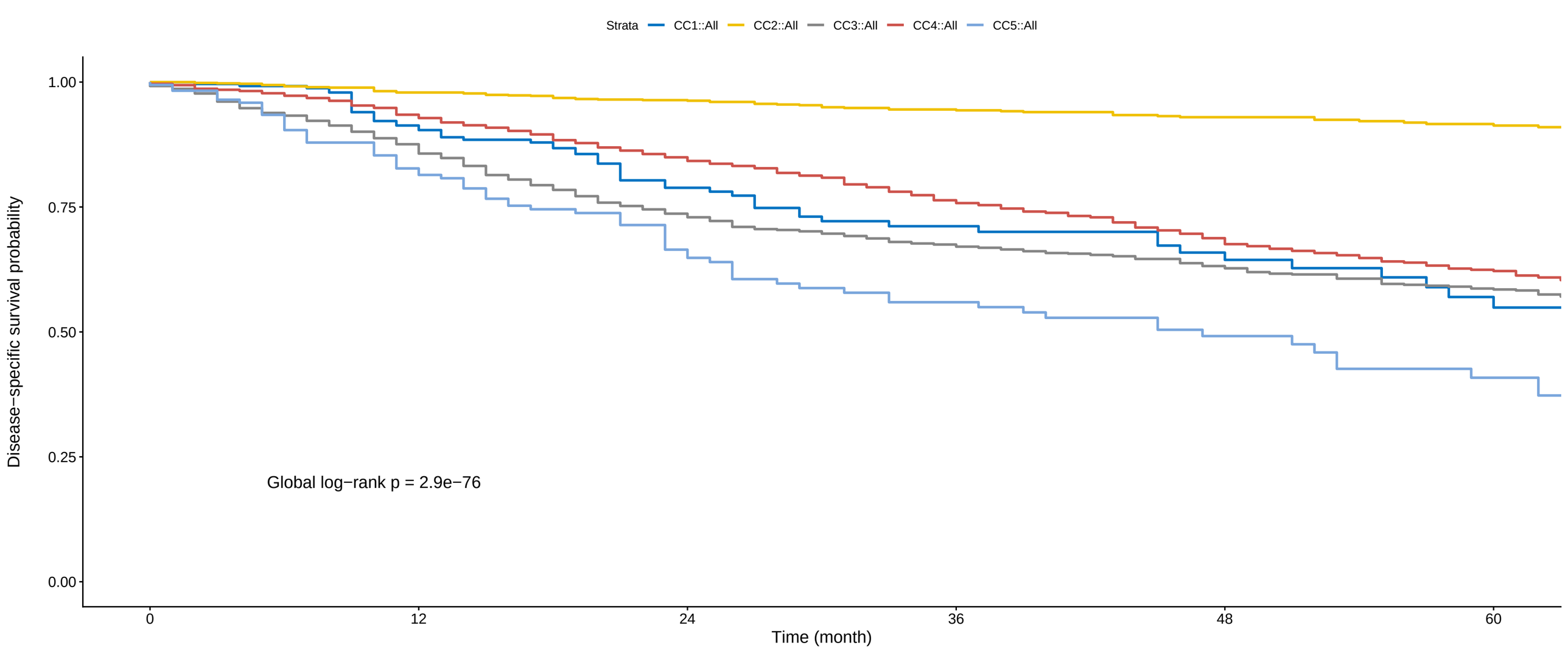
Number at risk

CC1::All	291	158	75	43	24	12
CC2::All	1240	1040	743	518	366	261
CC3::All	2100	1345	814	563	367	254
CC4::All	2748	1892	1117	701	500	353
CC5::All	183	100	54	42	24	16



Number at risk

CC1::All	177	102	44	22	13	6
CC2::All	690	629	449	291	204	153
CC3::All	915	767	501	357	244	172
CC4::All	1503	1211	731	465	329	231
CC5::All	46	40	23	17	11	8



Number at risk

CC1::All	280	199	103	66	45	27
CC2::All	1228	1083	809	573	412	299
CC3::All	2011	1501	915	633	424	308
CC4::All	2637	2091	1398	947	690	501
CC5::All	180	127	81	57	36	23